Analysis of Knee Joint Loading in Downhill Walking: A Linear Regression of Indoor Laboratory Data

TSM11 - Applied research methods WiSe 2025/26

Deutsche Sporthochschule Köln
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What do we know?- In the Lab

↑ Peak Tibiofemoral & Patellofemoral Compression Forces* DH > LG Walking (1)

Steeper inclination & Faster Speeds -> Higher lower extremety loading (2)

Controlled environment = High fidelity joint loading modelling (3)

1. Alexander, N., & Schwameder, H. (2016). *Gait & Posture*.
2. Schwameder, H. (2006). *Journal of Biomechanics*.
3. Strutzenberger, G., et al. (2022). *Gait & Posture*.

What do we know? - In the"wild"

+ hip–knee coordination*
+ gait variability (4)

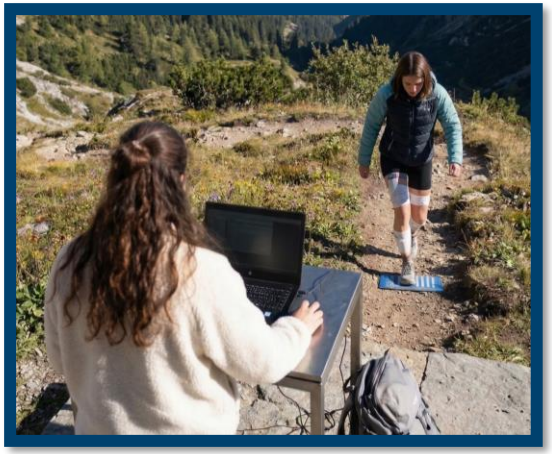
Tempo-parameters & 1-D Force data (5)

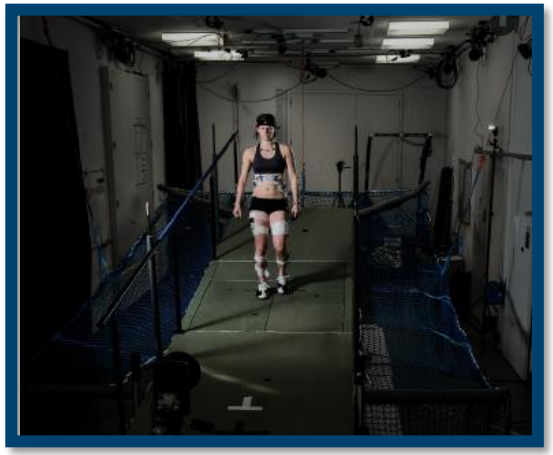
Remains unsolved to calculate joint moments & Forces (6)

4. Ippersiel, P., Shah, V., & Dixon, P. C. (2022). Gait & Posture.
5. Matikainen-Tervola, E., et al. (2025). Journal of Aging and Physical Activity.
6. Dixon, P., Shah, V., & Willmott, A. (2021). Gait & Posture

Introduction - The Two Environments

Ecological validity
Natural terrain
Full Kinematics



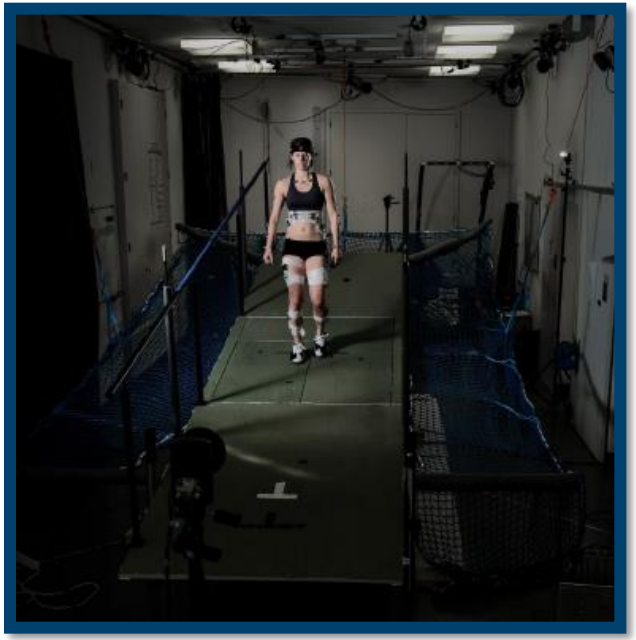


Gold Standard
Full Kinetics
Full Kinematics

Uncontrolled environment
Lower precision
Joint forces: ✗ unmeasurable

Artificial Gait
Limited Space

What do we don't know? - Research Gap



1 Can we predict peak joint loading?

H1: Peak knee joint loading can be accurately predicted using only kinematic and anthropometric variables.

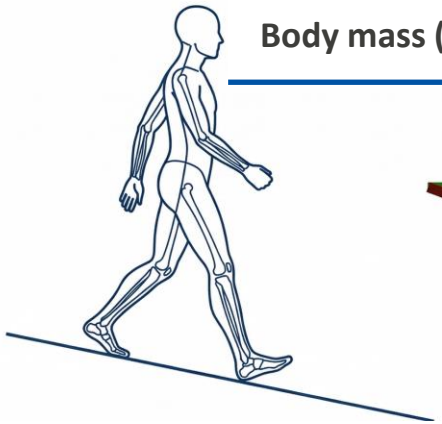
2 Which variables should we use?

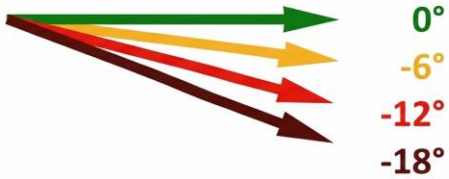
H1: Specific biomechanical variables (like joint angles, speed, and slope) are significant predictors of joint force.

3 How much does each value contribute?

Methods

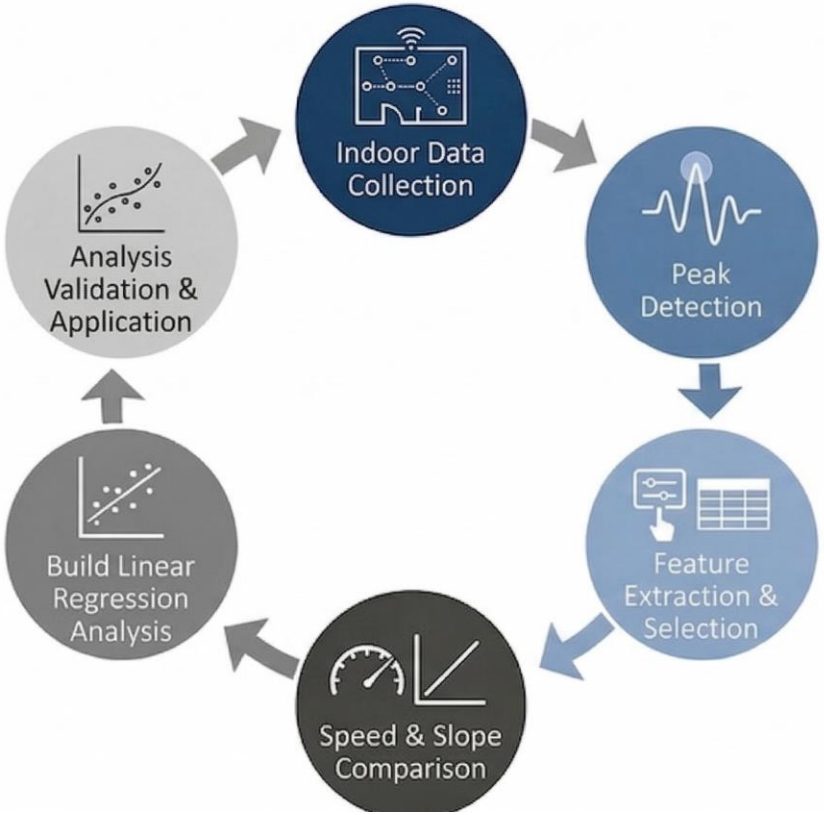
N	18 males	SD
Age (years)	27.0	± 4.9
Height (cm)	179.6	± 5.2
Body mass (kg)	74.5	± 8.4





- Controlled indoor slopes
- Constant speed of 1.2 $m \cdot s^{-1}$ or self-selected
- **Cross-Sectional, Secondary Lab Data Design**

Methods - Analysis

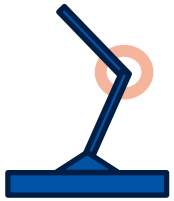


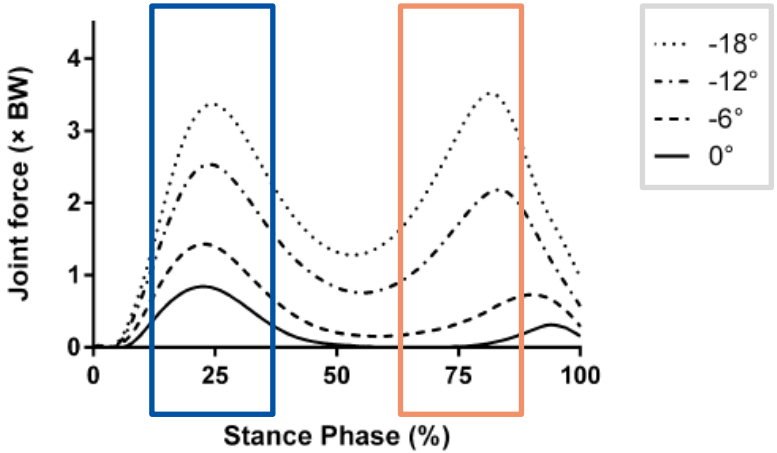
Details of Preprocessing

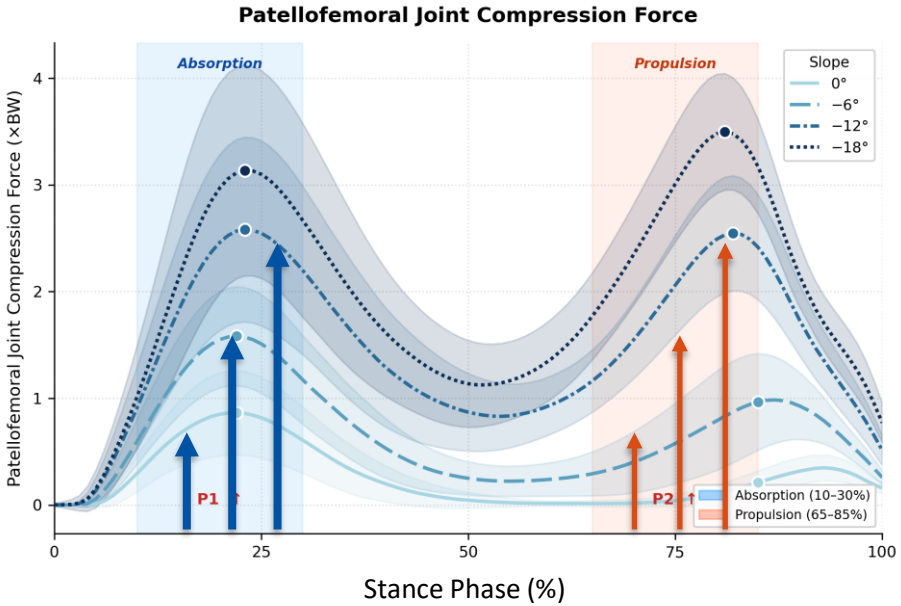
- Remove outiers
- Fixed participant ID as covariate

Results

Patellofemoral Compression Forces at different slope inclinations

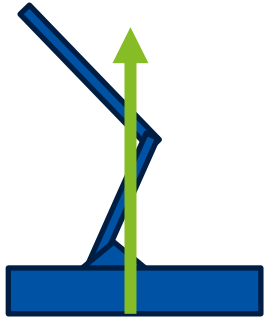


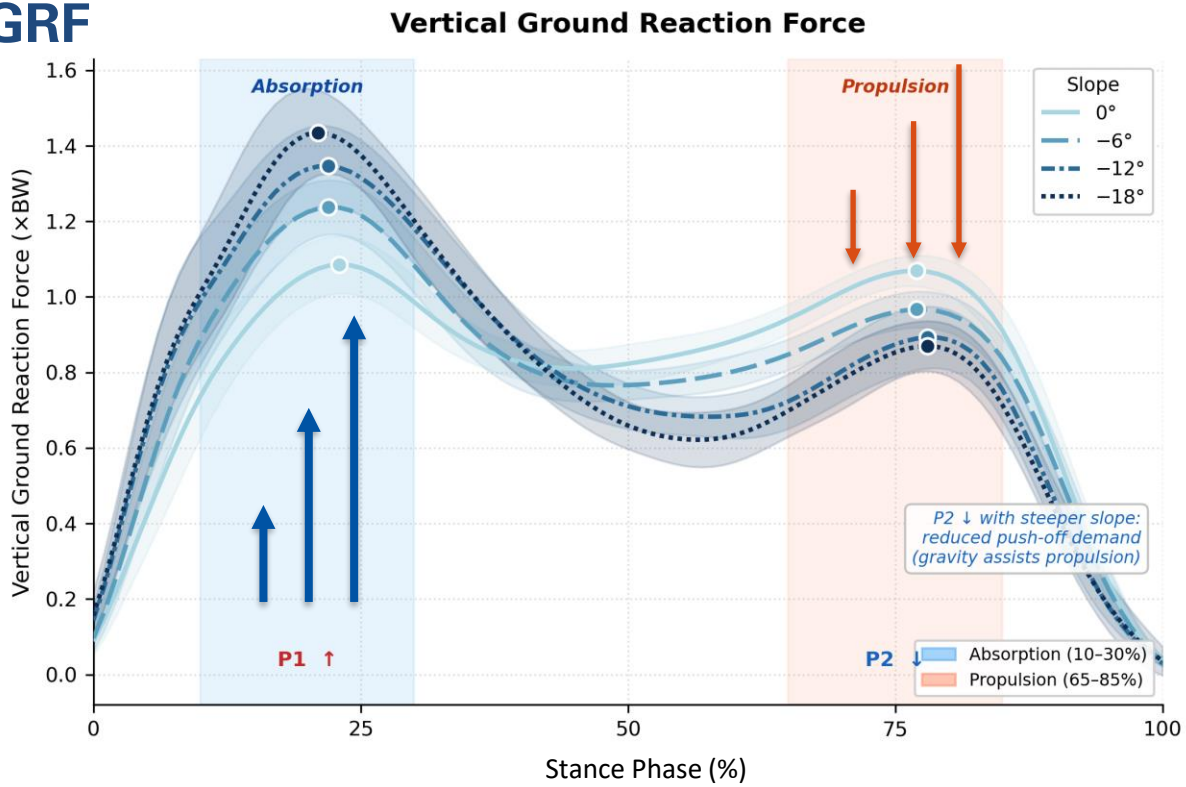




7. Alexander and Schwameder (2016)Gait & Posture

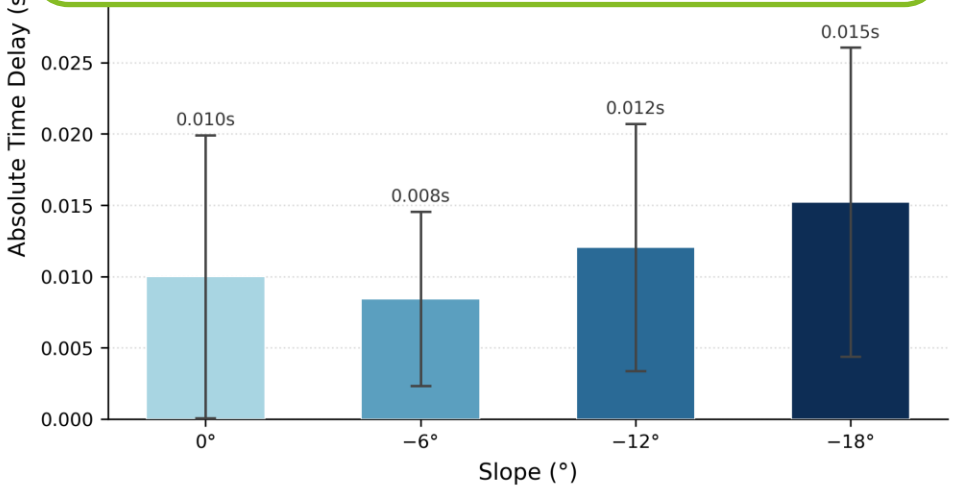
Results Vertical GRF



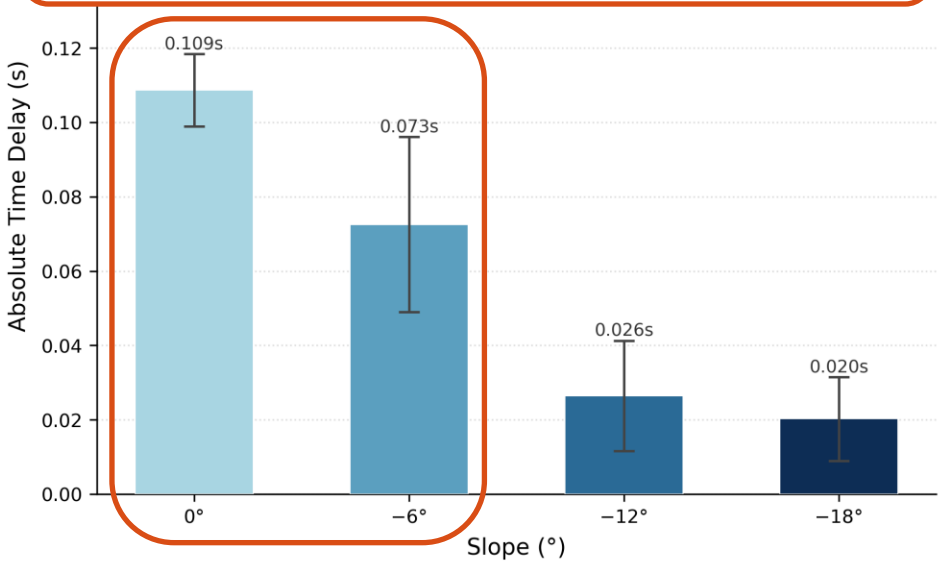


Absolute Time Delay between GRF Peak and PFc Peak by Slope and Phase | Excluded: [16, 20, 26]

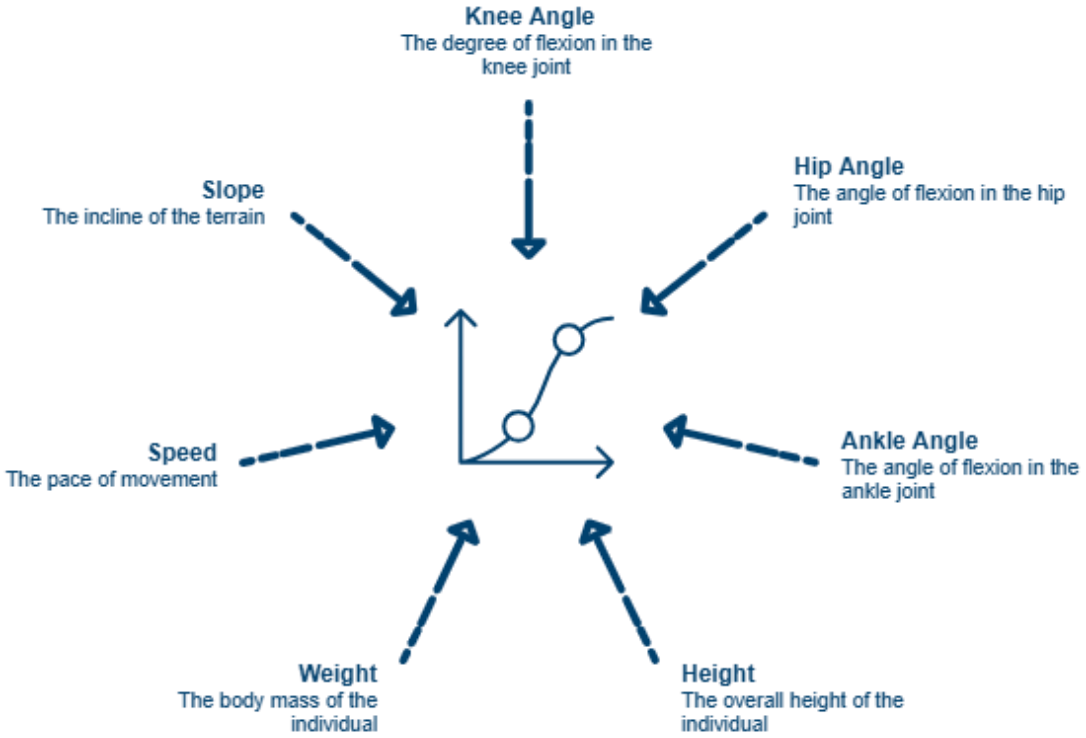
Because both peaks occur nearly simultaneously, we used the GRF peak as our timing reference.



Similar pattern at peak 2, but only when the incline is steep enough.



Building the Linear Regression



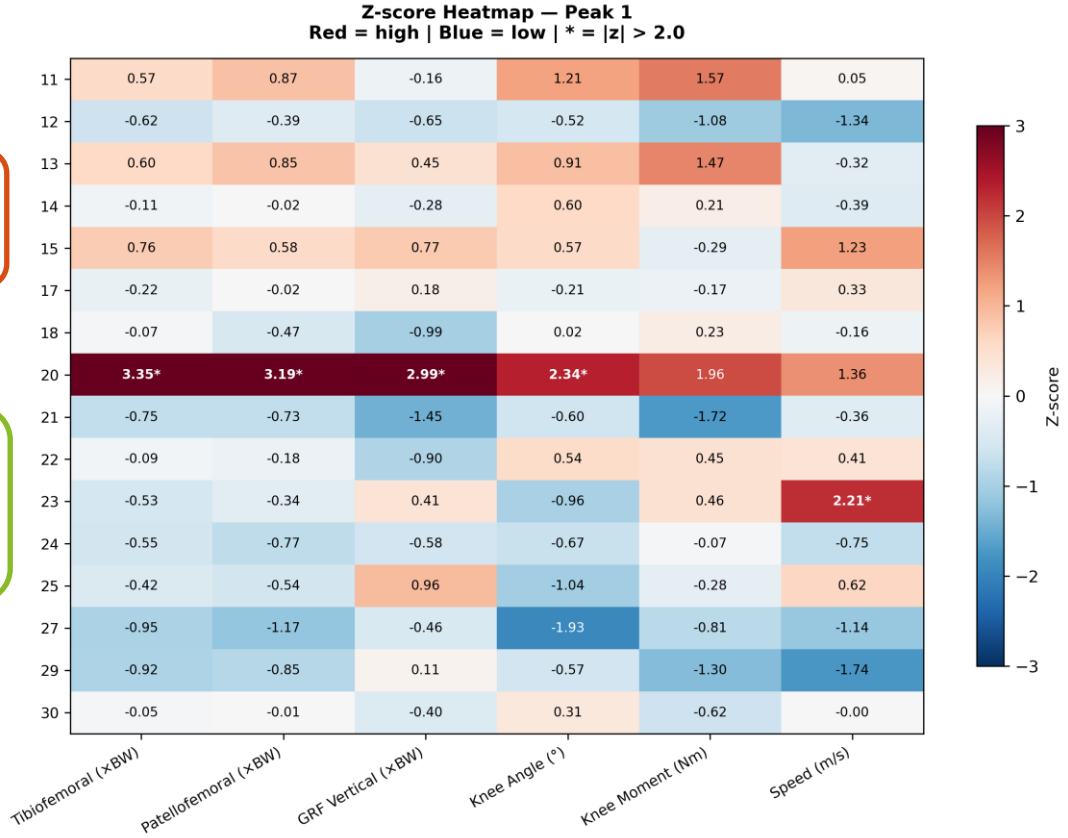
Feature selection

	Peak	Target	Scenario	R2	RMSE (%)	Variables Used
0	Peak 1	kneeFres	1. No Cadence	0.950	6.13%	angleKnee, angleHip, angleAnkle, height, weight, speed, slope
1	Peak 1	kneeFres	2. No Cadence & Ankle	0.950	6.15%	angleKnee, angleHip, height, weight, speed, slope
2	Peak 1	kneeFres	3. No Slope	0.944	6.50%	angleKnee, angleHip, angleAnkle, height, weight, speed, cadence
3	Peak 1	kneeFres	4. No Slope & Cadence	0.944	6.54%	angleKnee, angleHip, angleAnkle, height, weight, speed
4	Peak 1	kneeFres	5. All var	0.950	6.13%	angleKnee, angleHip, angleAnkle, height, weight, speed, slope, cadence
5	Peak 1	knee_PFc	1. No Cadence	0.954	11.33%	angleKnee, angleHip, angleAnkle, height, weight, speed, slope
6	Peak 1	knee_PFc	2. No Cadence & Ankle	0.953	11.37%	angleKnee, angleHip, height, weight, speed, slope
7	Peak 1	knee_PFc	3. No Slope	0.951	11.67%	angleKnee, angleHip, angleAnkle, height, weight, speed, cadence
8	Peak 1	knee_PFc	4. No Slope & Cadence	0.951	11.67%	angleKnee, angleHip, angleAnkle, height, weight, speed
9	Peak 1	knee_PFc	5. All var	0.954	11.27%	angleKnee, angleHip, angleAnkle, height, weight, speed, slope, cadence
10	Peak 2	kneeFres	1. No Cadence	0.668	13.48%	angleKnee, angleHip, angleAnkle, height, weight, speed, slope
11	Peak 2	kneeFres	2. No Cadence & Ankle	0.660	13.63%	angleKnee, angleHip, height, weight, speed, slope
12	Peak 2	kneeFres	3. No Slope	0.663	13.58%	angleKnee, angleHip, angleAnkle, height, weight, speed, cadence
13	Peak 2	kneeFres	4. No Slope & Cadence	0.662	13.60%	angleKnee, angleHip, angleAnkle, height, weight, speed
14	Peak 2	kneeFres	5. All var	0.669	13.47%	angleKnee, angleHip, angleAnkle, height, weight, speed, slope, cadence
15	Peak 2	knee_PFc	1. No Cadence	0.944	21.52%	angleKnee, angleHip, angleAnkle, height, weight, speed, slope
16	Peak 2	knee_PFc	2. No Cadence & Ankle	0.943	21.52%	angleKnee, angleHip, height, weight, speed, slope
17	Peak 2	knee_PFc	3. No Slope	0.944	21.50%	angleKnee, angleHip, angleAnkle, height, weight, speed, cadence
18	Peak 2	knee_PFc	4. No Slope & Cadence	0.943	21.52%	angleKnee, angleHip, angleAnkle, height, weight, speed
19	Peak 2	knee_PFc	5. All var	0.944	21.49%	angleKnee, angleHip, angleAnkle, height, weight, speed, slope, cadence

Two critrical points with participants

Z-score was calculated to remove outliers

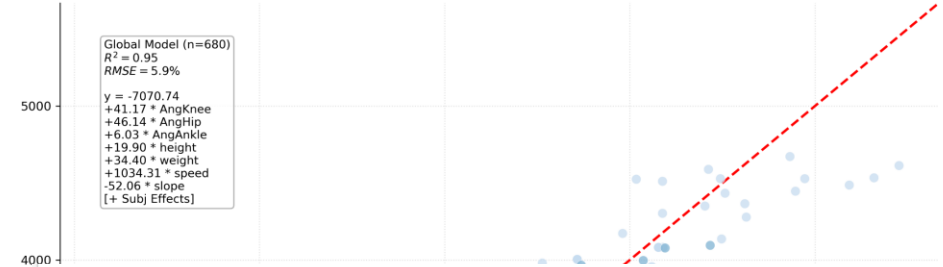
Fixed Subject ID as a covariate to decrease inter-subject differences



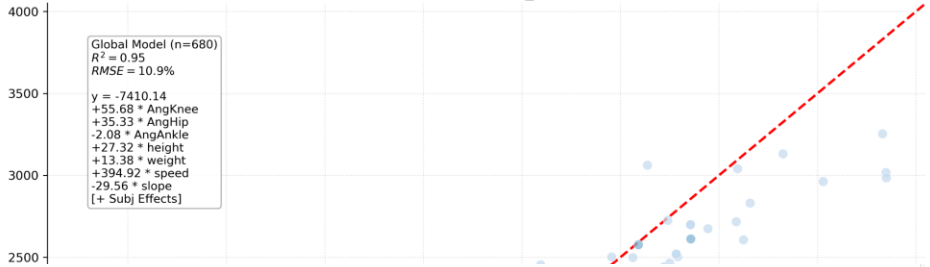
Global regression– peak 1

STEP 1: Global Regression Model (Peak 1)
(Participants 16, 20 & 26 Excluded)

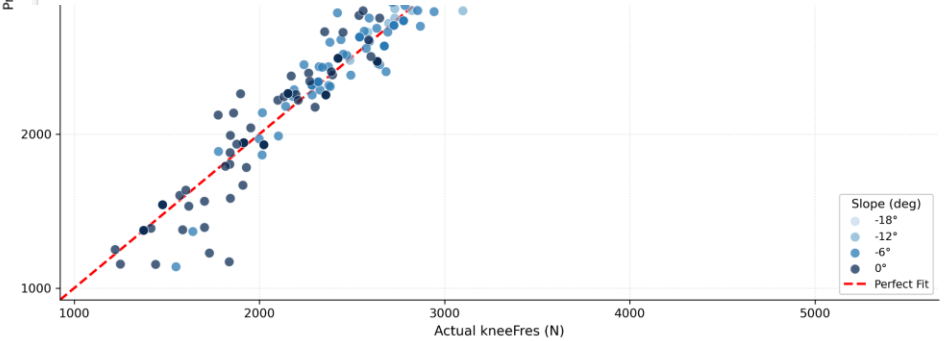
Target: kneeFres

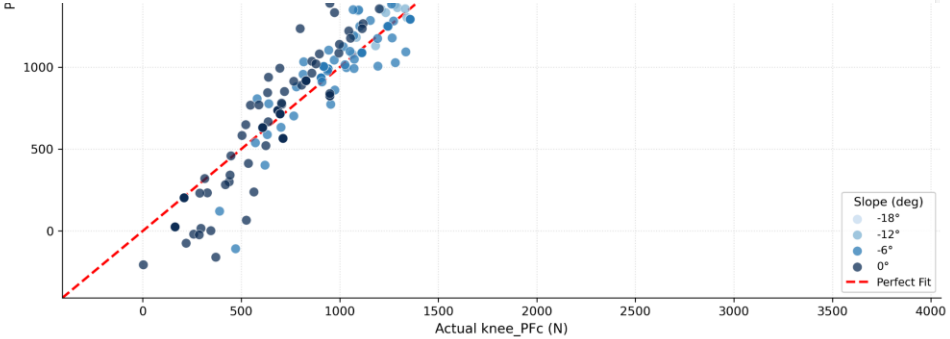


Target: knee_PFc



	Target	R^2	RMSE (%)	
0	kneeFres	0.950	5.92%	y = -7070.74 +41.17 * AngKnee +46.14 * AngHip +6.03 * AngAnkle +19.90 * height +34.40 * weight +1034.31 * speed -52.06 * slope
1	knee_PFc	0.953	10.87%	y = -7410.14 +55.68 * AngKnee +35.33 * AngHip -2.08 * AngAnkle +27.32 * height +13.38 * weight +394.92 * speed -29.56 * slope

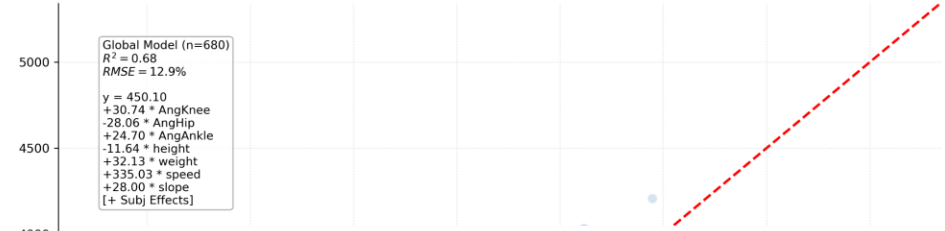




Global regression– peak 2

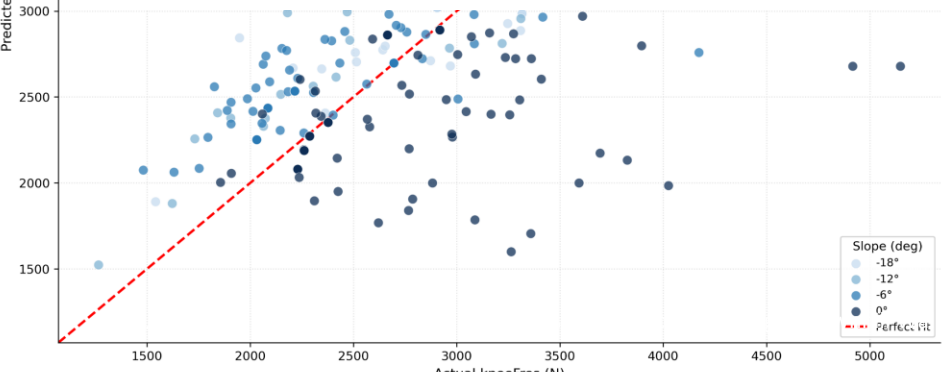
STEP 1: Global Regression Model (Peak 2)
(Participants 16, 20 & 26 Excluded)

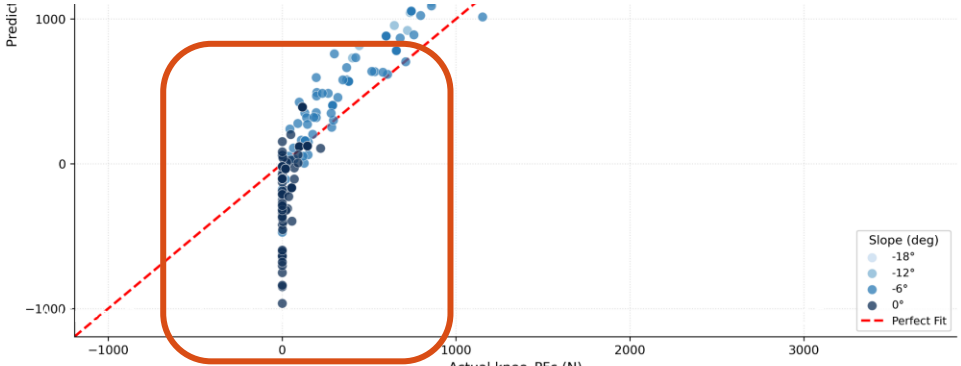
Target: kneeFres



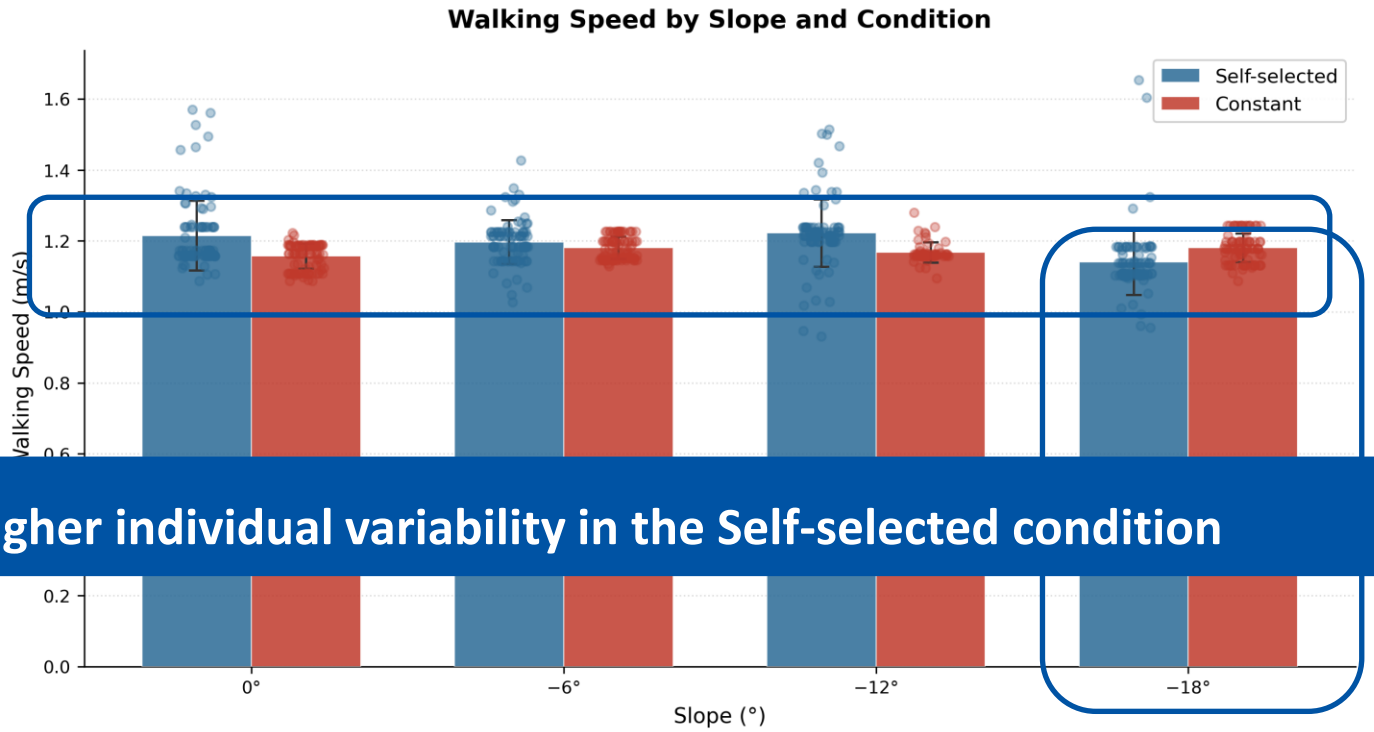
Target: knee_PFc

	Target	R^2	RMSE (%)	
0	kneeFres	0.677	12.91%	y = 450.10 +30.74 * AngKnee -28.06 * AngHip +24.70 * AngAnkle -11.64 * height +32.13 * weight +335.03 * speed +28.00 * slope
1	knee_PFc	0.951	19.68%	y = -1616.14 +49.67 * AngKnee -7.97 * AngHip -8.30 * AngAnkle +2.83 * height +9.69 * weight -56.42 * speed -10.55 * slope





Speed Comparison



Higher individual variability in the Self-selected condition

Discussion – Main Findings

Similar descriptive findings as Alexander & Schawemder (2016) supporting that preprocessing and modeling pipeline was correct.

Impact phase performance > Propulsion phase
R^2=0.95 against split results in Peak 2 R^2=0.68→0.95
Probably due to time misalignment between GRF and internal loading

Cadence was the only variable safely removed

Conclusion – Take Home Message

1. Knee Joint Compression Forces can be estimated from kinematics

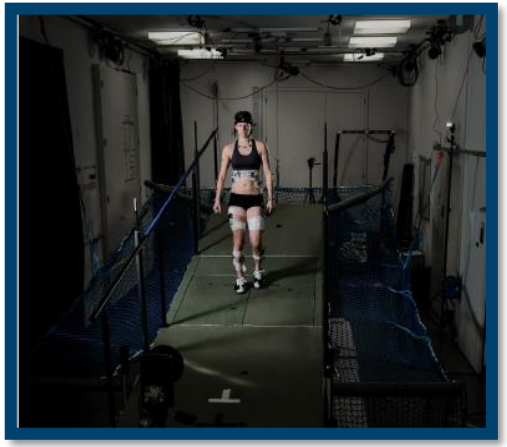
2. Impact Phase is more robust across more slopes

3. Sloped gait has its unique biomechanics and further research its needed

Limitations

1. Small Sample Power
2. No Muscle Activation
3. Temporal Misalignment
4. No female participants
5. Tibiofemoral compression force should be further studied

The Next Future Big Step



APPLY THE ANALYSIS TO THE REAL WORLD

Thank you for your attention!
Special thanks to Gerda Strutzenberger for her supervision and to Nathalie Alexander for providing the dataset

References

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